

KHQR Roll Sticker PDF Generator: Development & Verification Plan

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1. Overview and locked decisions

We build a browser-based tool that merges uploaded KHQR files with Excel rows and exports one vector-only PDF page per merchant sticker, about 3 weeks of work for one developer, deployed on Dokploy.

Inputs: many finished QR files (SVG), one Excel file, a background SVG, and a Bakong logo SVG.

Output: a PDF with one sticker per page, all fonts outlined, no raster images, in the page size chosen at export (Original, A3 to A7, custom), with or without bleed.

Topic	Decision
Template	Roll Sticker KHQR Single, artboard 317.5 × 427.5 pt, built exactly from the guide
QR	Uploaded QR is already correct. An SVG QR is embedded as provided. A raster QR (PNG or JPG) is redrawn as vector after a decode check. Placed at 134 × 134 pt
Bakong logo	Taken from the Logo upload slot and always scaled to 32 × 32 pt (a larger upload is scaled down, a smaller one up, uniformly), centered on the QR
Background	Uploaded SVG with no bleed. Fitted to the trim box only and never enlarged for the bleed. The bleed area is filled with a solid edge color
Merchant name	One Excel column, English and Khmer mixed. Max 25 characters per line, max 2 lines, wrapped and dropped by whole word. Nunito Sans ExtraBold for English, Nokora SemiBold for Khmer, 23 pt default, adjustable
MID	Line MID: <value> , Nunito Sans Regular, 10 pt default, adjustable. Max 15 characters, configurable for later growth

Topic	Decision
Export size	Page equals the chosen size exactly (A6 = 297.64 × 419.53 pt). Artwork scales uniformly, never stretched
Bleed	Choose bleed or no bleed at export. Default 3 mm, custom value editable, optional per-side values
Privacy	All processing in the browser. No files are uploaded to a server
Print	Download saves the PDF to your local computer through the browser, and Print opens the browser print dialog with the same generated PDF. Silent printing is not possible in a browser

2. Sticker layout specification (from the guide)

The artboard is 317.5 × 427.5 pt and every value below comes from your guide.

Coordinates are measured from the top-left corner and stored in `layout.json`, so a spec change is a data edit.

Element	Spec	Position (pt)
Corner frame	154.3 × 154.3 pt, 81.6 from left and right	x 81.6 to 235.9, y 98 to 252.3
QR	134 × 134 pt, 91.7 from left and right, 107.6 from top	x 91.7 to 225.7, y 107.6 to 241.6
Bakong logo	32 × 32 pt, centered on the QR	x 142.75 to 174.75, y 158.6 to 190.6
Name line 1	Nunito Sans ExtraBold (EN) or Nokora SemiBold (KH), 23 pt, centered, 25 characters max	Baseline at 299.9 pt (127.6 pt above the artboard bottom)
Name line 2	Same style, 11.8 pt gap below line 1	Only when the name wraps
MID	Nunito Sans Regular, 10 pt, centered, MID: <code><value></code> , 15 characters max	17 pt below the last name line

Rules for applying it:

- Gaps are measured between cap-height boxes (the blue lines in the guide), so text is positioned by cap height and baseline, not by font box.
- The background and bleed never change these positions, which are always relative to the trim box.
- The Affinity templates `qr-en-template.svg` and `qr-kh-template.svg` (same 0.743 ratio) are used as an overlay reference only. The 38 pt gap and the 127.6 pt bottom measure disagree by about 4 pt in the guide. We anchor on the measured baseline (299.9 pt), derive the other lines from the 11.8 pt and 17 pt gaps (about 327.9 pt for line 2 and 351.9 pt for MID in a two-line name), check the result against the vector templates, and record the final numbers in `layout.json`.

Fixed target sizes. The QR is always placed at 134 × 134 pt and the Bakong logo at 32 × 32 pt, whatever size the uploaded file has. Larger files are scaled down and smaller files are scaled up, using one uniform factor, so nothing is stretched.

3. Features and dashboard

The dashboard has five panels, and every setting is saved in the browser so a refresh keeps your work.

Panel	Controls
Assets	Upload slots for background SVG, Bakong logo SVG and multiple QR SVG files. Each slot shows a thumbnail, replace and reset. Built-in <code>a5.svg</code> , <code>bkb.svg</code> and <code>bkc.svg</code> are the defaults, and the black Bakong logo (bkb) is used unless you upload another
Data	Excel upload, sheet picker, header detection, column mapping (QR file name, Merchant name, MID), row range, table with warnings
Typography	Merchant name size (default 23 pt) and MID size (default 10 pt), each with a numeric field, slider and reset to guide value
Export	Page size (Original, A3, A4, A5, A6, A7, custom mm), Bleed or No bleed, bleed amount (default 3 mm, custom 0 to 10 mm, optional per side), bleed fill color (sampled by default), crop marks toggle, Download, Print

Panel	Controls
Preview	Live preview of the selected row with next and previous, plus an overlay of the guide's measurement lines

Behavior details:

- **QR matching:** each row finds its file by the mapped QR name. The list shows matched, missing and duplicate names, and a missing QR blocks that row with a clear message.
- **Upload validation:** QR files may be SVG, PNG or JPG: an SVG QR is embedded as is, and a raster QR is redrawn as vector (see section 5). Logo and background should be SVG, and a raster one is allowed only with a warning that it stays a raster image and fails the vector-only check.
- **Trimming:** over-limit cells are highlighted with a before and after view. Names drop whole words after line 2. MID is cut at 15 characters.
- **Mixed scripts:** English and Khmer in one cell are split into runs, each shaped with its own font and joined on one baseline.
- **Safe width:** a name line wider than the safe width (default: artboard width minus 20 pt each side) triggers a warning instead of a silent shrink, so the spec stays exact.
- **Persistence:** settings in localStorage, uploaded assets in IndexedDB.

Download to your computer. The Download button saves the file straight to your local machine, with no upload to a server.

- Default: one combined PDF with one sticker per page, named like `khqr-stickers-A6-bleed3mm.pdf` (size and bleed in the name).
- Option: a ZIP of single-sticker PDFs, one per row, each named by MID.
- Option: split a large batch into several PDFs of a chosen page count, for example 500 pages each.
- A progress bar shows while the file is built, and Cancel leaves no partial download.
- The file is created in the browser as a blob and saved through a normal download, so it lands in the browser's download folder. Chrome and Edge can also ask where to save it.

4. Architecture and tech stack

The app is a static single-page site that does all work in the browser, which keeps merchant data private and the server trivial.

Concern	Choice
App	Vite, React, TypeScript, Tailwind CSS (React is the recommended choice; Vue only if your team prefers it, decided before Phase 1)
Excel	SheetJS (<code>xlsx</code>)
Text shaping	<code>harfbuzzjs</code> (correct Khmer clusters, stacked subscripts, vowel reordering)
Segmentation	<code>Intl.Segmenter</code> for words and grapheme clusters (<code>km</code> and <code>en</code>)
Fonts	Nunito Sans ExtraBold and Regular, Nokora SemiBold, all OFL, bundled as TTF
SVG handling	<code>svgo</code> for cleanup, a path parser for glyph outlines and QR validation, and a QR reader (<code>zxing-wasm</code> or <code>jsQR</code> , chosen in the Phase 3 spike) to decode raster QR codes and rebuild them as vector
PDF	<code>pdf-lib</code> plus a custom SVG-to-PDF converter (or <code>svg-to-pdfkit</code> , decided in the Phase 2 spike), with <code>TrimBox</code> and <code>BleedBox</code> set explicitly. No fonts or images embedded
Batching	Web Worker builds pages and streams them to the PDF, target 1,000+ stickers without freezing the tab
State	Zustand store, <code>localStorage</code> for settings, <code>IndexedDB</code> for assets
Tests	Vitest, Playwright, <code>poppler-utils</code> (<code>pdffonts</code> , <code>pdfimages</code> , <code>pdftoppm</code>), <code>pixelmatch</code> , <code>zbar</code> for QR decoding
Hosting	Docker (Node build, <code>nginx</code> serve) on Dokploy

Data flow for one export:

1. Read settings, assets and the Excel rows.
2. For each row, match the QR file and normalize the name and MID (trim, wrap, limit).
3. Shape each text run with HarfBuzz and turn glyphs into path outlines.
4. Compose the sticker group at guide coordinates: background, frame, QR, logo, name lines, MID.

5. Apply the page transform: uniform scale, centered on the trim box, background cover scaled to the bleed box.
6. Write the page into the PDF with its page boxes, then stream the next row.
7. Offer the file for download, or load it into a hidden iframe and call `print()`.

5. Core algorithms

Text outlining. The name is split into script runs (Latin, Khmer). Each run is shaped by HarfBuzz with its font, and each glyph outline becomes an SVG path with the shaped offsets applied. The runs are joined on one baseline and centered as a single line. No `<text>` element and no embedded font reaches the PDF.

Name wrapping (max 2 lines, 25 characters, by word).

1. Split into words with `Intl.Segmenter` (spaces for Latin, dictionary segmentation for Khmer).
2. A line fits when it has 25 or fewer grapheme clusters and its outlined width is within the safe width.
3. Fill line 1 with whole words, then line 2 the same way.
4. Words that do not fit on line 2 are dropped. No word is cut and no ellipsis is added.
5. Only a single word longer than 25 clusters is cut, on a grapheme boundary, so Khmer glyphs never break.
6. Every changed row is flagged in the table with original and result.

MID. Format `MID: <value>`, trimmed to the configured limit (15 by default, `limits.mid`), no wrapping.

Vertical flow. Line 1 starts at the guide offset below the QR. Line 2 follows with the 11.8 pt gap and MID sits 17 pt below the last name line. A one-line name therefore moves MID up by one line pitch. If you prefer MID at a fixed position instead, it is one config switch (`mid.anchor`).

Background and bleed. The background is fitted to the trim box with one uniform scale and centered, so it covers the trim area and is never enlarged for the bleed. In bleed mode, the extra area around it is filled with a solid color, sampled per side from the background's edge by default or chosen in the export panel. In no-bleed mode only the trim area is exported. The trim area looks identical in both modes. A flat edge color works best, and a pattern or gradient at the edge triggers a warning because the fill will not match it.

Page size and bleed.

Mode	A6 page	Page boxes
No bleed	297.64 × 419.53 pt	MediaBox equals TrimBox
Bleed, default 3 mm (8.50 pt)	314.65 × 436.54 pt	TrimBox is exact A6, BleedBox equals MediaBox

The sticker artboard (317.5 × 427.5 pt) is scaled uniformly with $s = \min(\text{trimW} / 317.5, \text{trimH} / 427.5)$ and centered in the trim box. A6 scales it to about 93.7%, and nothing is stretched. Optional crop marks add margin outside the bleed.

Logo and QR. The logo SVG is placed at 32 × 32 pt over the QR center with no extra ring, because the supplied logo already includes the white ring. This is confirmed against the template overlay in Phase 2. An SVG QR is embedded as given, and a raster QR is redrawn as vector as described next.

Scaling uploaded QR and logo. The app reads the file's size from the SVG viewBox (falling back to width and height in any unit, converted to pt) or from the pixel size of a raster file. It then applies one scale factor, $s = \text{target} / \max(\text{width}, \text{height})$, equally in X and Y, and centers the artwork in its target box: 134 × 134 pt for the QR and 32 × 32 pt for the logo.

- A square file fills its box exactly, whether the upload is larger or smaller than the target.
- A non-square file is fitted inside the box without cropping or stretching, and gets a warning because a QR must be square.
- Empty padding around the logo is ignored, so the visible logo measures 32 pt. The QR is measured on its dark module area, with the quiet zone excluded, because the 134 pt in the guide is the visible code. This is confirmed against the template overlay in Phase 2.
- Transforms and clip paths inside the upload are applied before measuring.
- The preview and table show the original size and the scale applied, for example "scaled from 512 × 512 to 134 × 134 pt".

Raster QR redraw (PNG or JPG). If an uploaded QR is a raster image, it is rebuilt as clean vector before it is placed, so the exported PDF still contains no images.

1. Decode the image with a QR reader and keep the payload, version and error-correction level.
2. Find the code's corners, correct rotation and perspective, and detect the module grid (for example 29 × 29 modules for version 3, plus the quiet zone).
3. Sample the center of every module and threshold it, which gives the exact black and white matrix of the uploaded image.

4. Draw the dark modules as one merged path of rectangles, so no hairline seams show between modules, on a square grid scaled to 134 × 134 pt.
5. Verify: render the new vector, decode it, and require the same payload as the original, also with the Bakong logo in place. If the check fails, rebuild from the decoded payload with the same version and error-correction level, which scans the same but may use a different mask pattern. If that also fails, refuse the file and say why (blurry, cropped, too low resolution or damaged).

Rules for raster QR files:

- Resolution: warn below about 4 px per module and refuse below 2 px per module.
- The image must be square, and the quiet zone is kept.
- A QR with a logo already baked in is warned about, because the modules under it are lost.
- Rows using a redrawn QR carry a "redrawn" badge in the data table and the preview.
- The export panel has a setting "Redraw raster QR as vector", on by default. Switched off, a raster QR is refused.
- An SVG QR is always preferred when you have one, because a redraw is a reconstruction of the image.

6. Repository structure and config model

```
khqr-sticker/
  Dockerfile  nginx.conf  docker-compose.yml  .dockerignore
  public/fonts/      NunitoSans-ExtraBold.ttf  NunitoSans-Regular.ttf
  Nokora-SemiBold.ttf
  public/assets/      a5.svg  bkb.svg  bkc.svg  (defaults)
  src/
    config/layout.json      all guide numbers
    config/limits.json      name 25, lines 2, mid 15, safeMargin 20
    core/text/              shape.ts  outline.ts  runs.ts  wrap.ts  trim.ts
    core/layout/            sticker.ts  page.ts  background.ts
    core/pdf/               writer.ts  boxes.ts
    core/svg/               parse.ts  validate.ts  embed.ts
    core/excel/             read.ts  mapColumns.ts
    workers/export.worker.ts
    ui/                     AssetsPanel  DataPanel  TypographyPanel
  ExportPanel  Preview
  store/              settings.ts  assets.ts
  tests/              unit  integration  e2e  fixtures  golden
  tools/              preflight.sh  overlay-diff.ts  qr-scan.sh
```

Settings stored per browser:

Key	Default	Notes
<code>nameSizePt</code>	23	Merchant name font size
<code>midSizePt</code>	10	MID font size
<code>limits.nameChars</code>	25	Per line
<code>limits.nameLines</code>	2	Fixed by the guide decision
<code>limits.mid</code>	15	Raise for longer MIDs later
<code>safeMarginPt</code>	20	Warn when a line exceeds artboard width minus 2 × margin
<code>bleedMm</code>	3	Custom 0 to 10, optional per side
<code>bleedEnabled</code>	false	Chosen at export
<code>pageSize</code>	original	Original, A3 to A7, custom
<code>cropMarks</code>	false	Adds margin outside the bleed

7. Phased development plan

Seven phases take about 3 weeks. A phase is finished only when its exit criteria pass, and the highest-risk work (Khmer outlining and exact layout) comes first.

Phase	Tasks	Deliverable	Exit criteria	Est.
1. Foundation	Scaffold Vite + React + TS, lint and test setup, Dockerfile, nginx, first Dokploy deploy	Empty app live on a Dokploy URL	Deploy from <code>main</code> works and the health check passes	1 day
2. Outline and layout core	HarfBuzz integration, glyph-to-path, script runs, mixed EN and KH line, <code>layout.json</code> , sticker composer, overlay against the two Affinity templates	A script that renders a sample sticker to SVG and PDF	Overlay diff under 0.1 pt on all guide elements. Khmer samples shape correctly	3 to 4 days

Phase	Tasks	Deliverable	Exit criteria	Est.
3. Dashboard and assets	Panel shell, upload slots for background, logo and QR, SVG validation, raster QR vectorizer with decode check, IndexedDB and localStorage persistence	Working asset management	Refresh keeps assets. Bad files refused with clear messages	2 days
4. Data and typography	Excel read, sheet and column mapping, QR matching, wrap and trim engine, font size controls, table warnings, live preview	End-to-end preview of any row	Wrap and trim tests pass. Preview equals export geometry	3 days
5. Export	PDF writer with TrimBox and BleedBox, size presets, bleed options, background fit and bleed edge fill, crop marks, download and print flow	Exact-size PDF for each preset	A6 page equals 297.64 × 419.53 pt. Bleed boxes correct. Preflight passes	3 days
6. Hardening	Worker batching, progress bar and cancel, memory checks, preflight suite in CI, QR scan tests, visual regression	Green test suite and performance report	1,000 rows export without freezing. All checks in section 8 pass	2 days
7. Release	UX polish, error messages, user guide, production deploy, backup of the settings export	Production site	Acceptance checklist in section 8 signed off	1 day

Working rules for the build in Claude Code:

- Phase 2 starts as a spike with a single sample sticker before any UI is built.
- Each phase ends with a commit tagged `phase-N`, and its tests run in CI before the next phase starts.
- Anything that changes a number in the guide goes through `layout.json` and a note in the repo changelog.

8. Verification plan

Every requirement has an automated check that fails the build, plus a short manual acceptance pass before release.

8.1 PDF preflight (runs on every exported PDF in CI)

Check	Tool	Pass condition
No fonts	<code>pdffonts</code>	Lists zero fonts
No raster images	<code>pdfimages -list</code>	Lists zero images
Vector only	Content stream scan with <code>qpdf --qdf</code>	Only path, fill, stroke, clip, transform and graphics-state (opacity) operators. No text operators (<code>BT</code> , <code>Tj</code>) and no image <code>Do</code>
Page size	<code>pdfinfo -box</code>	A6 no bleed: 297.64 × 419.53 pt within 0.01 pt. A6 with 3 mm bleed: 314.65 × 436.54 pt
Boxes	<code>pdfinfo -box</code>	TrimBox equals the requested size. BleedBox equals MediaBox in bleed mode, and MediaBox equals TrimBox in no-bleed mode
Page count	<code>pdfinfo</code>	Equals the number of exported rows
Scaling	Geometry test	One uniform scale factor in X and Y, within 1e-6

8.2 Layout accuracy

Check	Method	Pass condition
Guide positions	Unit test on <code>layout.json</code> against the guide table	QR 134 pt, logo 32 pt, offsets 91.7, 107.6, 81.6, 98 match exactly
Template overlay	Render at 600 DPI, diff against <code>qr-en-template.svg</code> and <code>qr-kh-template.svg</code> with <code>pixelmatch</code>	Edge deviation under 0.1 pt, and the two documented guide inconsistencies recorded
Gaps	Measure cap-height boxes in the output	11.8 pt between name lines, 17 pt to MID
Font sizes	Compare outline height to font metrics	23 pt and 10 pt by default, and changes follow the dashboard values
Visual regression	Golden PNGs at 300 DPI for 8 reference stickers	Pixel difference under 0.1%

8.3 Text engine tests

Case	Expectation
Name of exactly 25 characters	One line, no change
Name of 26 characters with several words	Wraps to line 2 by word
Name long enough to exceed 2 lines	Extra words dropped, none cut, flagged in the table
Single word over 25 clusters	Cut on a grapheme boundary
Khmer name with stacked subscripts and vowel signs	Shapes correctly, never splits a cluster, matches a HarfBuzz reference render
Mixed English and Khmer in one cell	Two fonts on one baseline, centered as one line
Wide letters (<code>www</code> , <code>mmmm</code>) at 25 characters	Safe-width warning shown, no silent shrink
MID of 15, 16 and 20 characters	15 kept, others cut and flagged, limit changeable through config
Empty, whitespace-only and emoji cells	No crash, row flagged

8.4 Data and upload tests

- Excel: `.xlsx` and `.xls`, a sheet with a header offset, empty rows, duplicate QR names, a missing QR file, 5,000 rows.
- Uploads: a PNG or JPG QR is redrawn as vector and verified, a blurry, cropped or low-resolution raster QR is refused with a reason, a non-square QR is warned, an SVG with embedded `<image>` is rejected for QR and warned for background and logo.
- Persistence: reload keeps settings and assets, reset restores the guide defaults.

Scaling tests. Upload QR files at 50, 134, 500 and 2,000 pt and logo files at 16, 32 and 200 pt, using viewBox only, mm units and px units. The placed QR must measure 134 pt and the logo 32 pt within 0.01 pt, with the aspect ratio unchanged and both centered within 0.01 pt. A non-square upload must show the warning.

8.5 QR integrity

Decode every sample QR with `zbar` before and after export, rasterized at A6 and A7 from the PDF, with the logo in place. The payload must be identical. A failure at A7 produces a warning in the export panel.

Raster QR test set. Use PNG and JPG versions of the sample QR files at 4, 6 and 10 px per module, rotated, slightly skewed and with JPEG compression. Each rebuilt vector must decode to the same payload, with the logo in place, at A6 and A7. Blurry, cropped and 2 px per module images must be refused with a reason and never exported.

8.6 Performance and stability

Test	Pass condition
1,000 stickers exported	Under 60 seconds on a mid-range laptop, UI stays responsive
Memory	Peak below 1.5 GB, and the PDF streams page by page
Cancel	Cancel stops the worker within 1 second and leaves no partial download
Browsers	Latest Chrome, Edge and Firefox, plus Safari for the preview and download

8.7 Manual acceptance checklist

- ☐ Upload background, logo, QR files and Excel, map columns, preview matches the guide image
- ☐ Change merchant name size and MID size, preview and PDF both follow
- ☐ Export A6 without bleed, open in Illustrator or Acrobat, confirm 297.64×419.53 pt and that text is outlines
- ☐ Export A6 with 3 mm bleed and with a custom bleed, confirm Trim and Bleed boxes
- ☐ Export A7, A5, A4 and A3, confirm uniform scaling with no stretching
- ☐ Print through the browser dialog to a real printer, confirm size at 100% scale
- ☐ Print one test sheet on the actual roll sticker printer and scan the QR with a Cambodian banking app
- ☐ Khmer names reviewed by a Khmer reader for correct shaping
- ☐ Click Download, confirm the file lands on the computer, opens in a PDF viewer, and the ZIP and split options produce the right files and names

9. Dokploy deployment

The site is a static build served by nginx, so Dokploy needs no database, volume or environment secrets.

1. **Image:** a multi-stage Dockerfile. Stage one is `node:20-alpine` and runs `npm ci` and `npm run build`. Stage two is `nginx:alpine` serving `dist/` with a custom `nginx.conf`.
2. **nginx settings:** single-page fallback to `index.html`, gzip, long cache for hashed assets, no cache for `index.html`, correct MIME types for `.wasm` (HarfBuzz) and `.ttf`, and security headers (`X-Content-Type-Options`, `Referrer-Policy`, a CSP that allows `blob:` plus workers and WebAssembly).
3. **Dokploy app:** create an Application from the Git repository, build type Dockerfile, container port 80.
4. **Domain:** add the domain in Dokploy with HTTPS through Traefik and Let's Encrypt.
5. **Pipeline:** auto-deploy on push to `main`. A `develop` branch deploys to a staging app for review before production.
6. **Health check:** an HTTP check on `/` in Dokploy, plus a `/healthz` location in nginx returning 200.
7. **CI before deploy:** lint, unit tests, build, then the preflight suite from section 8, so a broken export never reaches production.
8. **Rollback:** Dokploy keeps previous deployments, and each release is tagged in Git (`v1.0.0`) so any version can be redeployed.

No user data is stored on the server. If you later add features that need storage, such as saved templates shared between users, that would be a separate backend phase.

10. Risks, assumptions and handoff

The biggest risk is Khmer text shaping, so it is built and tested first in Phase 2. The second risk is SVG-to-PDF fidelity: the Affinity files use clip paths, nested transforms and 80% opacity, so Phase 2 also includes a conversion spike that compares the PDF with the SVG render before any UI is built. The third risk is the raster QR redraw: a blurry or low-resolution image cannot be rebuilt reliably, so every redrawn QR must decode to the same payload as the original or it is refused.

Risk	Impact	Mitigation
Khmer shaping errors (subscripts, vowel order)	Broken names on stickers	HarfBuzz shaping, reference renders, review by a Khmer reader
Bleed fill does not match a patterned background edge	Slight color difference in the bleed area if the print shop cuts off-trim	Edge color sampled per side, color picker in the export panel, warning for patterned edges
Guide inconsistency (38 pt vs 127.6 pt, about 4 pt)	Slight vertical offset	Follow the Affinity vector templates, record final values in <code>layout.json</code>
QR too small at A7	Scan failure	Automatic scan test and a warning in the export panel
Very large batches	Slow or frozen tab	Worker, streaming PDF writer, progress and cancel
Uploaded SVG with hidden raster images	Breaks the vector-only rule	Validation rejects or warns, preflight confirms
Silent printing expected	Browser always shows a print dialog	Documented. A CUPS or IPP service is an optional later phase

Assumptions to confirm:

- The background SVG supplied (`a5.svg` or your own) has no bleed and a proportion close to the sticker.

- Excel has one name column (English and Khmer mixed), a QR file name column and an MID column.
- A name wider than the safe width is warned about, not shrunk.
- When the name is one line, MID moves up by one line pitch instead of staying at a fixed position.

Before development starts in Claude Code, please confirm this plan and send:

- ☐ Approval of this plan, or the changes you want
- ☐ The three font files, or permission to download the OFL versions
- ☐ A sample Excel file with real column names and a few Khmer names
- ☐ A few sample QR SVG files, and your final background and Bakong logo files